DARUN EISWAR S

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B.Tech Electronics and Communication Engineering

Professional Summary

Electronics and Communication Engineering graduate with hands-on experience in C programming, embedded systems, and Linux-based platforms (Raspberry Pi), plus hardware-software integration across FPGA-based SoC designs using AMBA AHB/APB and AXI protocols. Skilled in protocol-level debugging, RTL validation, and building real-time embedded applications. Seeking an Embedded Systems / Firmware Engineer role focused on hardware-software integration on Linux-based embedded platforms.

Technical Skills

- **Embedded Systems & Programming:** C, Python, Embedded Systems, Linux, Raspberry Pi
- **Hardware-Software Integration:** AMBA AHB/APB, AXI4-Lite, PS-PL Communication (Xilinx Zynq), Protocol-Level Debugging
- **FPGA/RTL Design:** Xilinx FPGA, Vivado, Verilog, SystemVerilog, VHDL, RTL Design, Timing Analysis, Synthesis
- **Debugging & Validation:** RTL Simulation, Waveform Debugging, Protocol Verification, Hardware Validation
- **Tools & Version Control:** Git, MATLAB, LTspice

Experience

Research Intern – FPGA Systems *National Institute of Technology (NIT), Trichy* | *Apr 2025 – Jun 2025*

- Designed and validated RTL modules for an FPGA-based Network-on-Chip using Verilog/SystemVerilog.
- Performed synthesis, implementation, timing analysis, and hardware validation in Vivado.
- Implemented and tested AXI-based communication between Processing System and Programmable Logic on a Xilinx Zynq FPGA, directly involving hardware-software integration.

VLSI Design Intern *Maven Silicon* | *Jul 2024 – Aug 2024 (Virtual)*

- Designed and verified an AMBA AHB-to-APB bridge using SystemVerilog.
- Performed RTL simulation, waveform debugging, and protocol verification.
- Worked with RTL design, finite state machines (FSMs), and AMBA bus protocols.

Projects

AI-Based Driver Monitoring System *Python, Raspberry Pi (Linux), OpenCV* | *Jun 2024 – Oct 2024*

- Built a real-time embedded vision system on Linux-based Raspberry Pi, improving driver fatigue detection accuracy by approximately 8–10% using EAR, MAR, head pose, and blink frequency analysis.
- Reduced inference latency by approximately 30–35% and increased processing speed from 10 FPS to 15 FPS through embedded model optimization on resource-constrained hardware.
- Reduced false drowsiness alerts by approximately 15–20% using adaptive fatigue scoring and event validation.

Intelligent QoS-Aware Network-on-Chip for AI Accelerators *SystemVerilog, Vivado* | *May 2025 – Sept 2025*

- Reduced average packet latency by approximately 15% using QoS-aware adaptive routing.
- Reduced network congestion by approximately 18% and improved throughput by approximately 12% using dynamic virtual channel allocation.
- Validated the complete NoC on a Xilinx FPGA at 100 MHz through timing, latency, throughput, and resource analysis.

Adaptive Mixed-Precision Transformer Inference Accelerator on FPGA *SystemVerilog, Vivado* | *Dec 2025 – Mar 2026*

- Reduced DSP and BRAM utilization by approximately 15% using layer-wise adaptive accumulator bit-width optimization.
- Improved inference throughput by approximately 12% using a runtime-configurable mixed-precision computation pipeline.
- Validated the accelerator on a Xilinx FPGA at 100 MHz while evaluating latency, resource utilization, power, and timing closure.

Education

Amrita Vishwa Vidyapeetham *2022 – 2026*
B.Tech Electronics and Communication Engineering, CGPA: 8.39/10

Holy Angels Matric Hr. Sec. School *2021 – 2022 / 2019 – 2020*
Class XII – 92.8% | Class X – 89.4%

Achievements & Certifications

- Published research paper: *Dyn MM: A Dynamic Modulus-Update Memory-Based Modular Multiplier on FPGA*, IETE Journal of Research (2026).
- Research Internship Certificate – NIT Trichy; VLSI Digital Internship Certificate – Maven Silicon.

Soft Skills & Languages

Problem Solving, Technical Communication, Hardware Debugging, Team Collaboration | Tamil (Native), English (Professional), Hindi (Limited Working)